RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering

Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

UNIT I (Information Theory)

Degree, Semester & Branch: V Semester B.E. ECE A
Course Code & Title: EC8501 Digital Communication
Name of the Faculty member: Mr.P.Gunasekaran
Name of the Topic: Shannon Fano Coding
Name of the Innovative Practice: Simulation
Date & Duration: 11.07.2019 & 30 Minutes
ICT Tool Used: Computer, Matlab Software

Description:

MATLAB (matrix laboratory) is a multi-paradigm numerical computing environment and proprietary programming language developed by Mathworks. MATLAB allows matrix manipulations, plotting of functions and data, implementation of algorithms, creation of user interfaces, and interfacing with programs written in other languages.

Goals (Learning Outcomes):

To enable the students to learn the fundamentals of Shannon fano coding and some of the matlab commands.

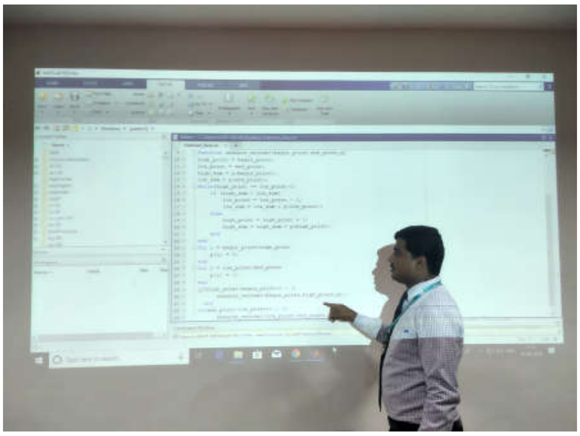
Use of appropriate methods:

Justification for choosing the topic using Hardware Demonstration

The simulation can help the students to solve the problems in an effective way. They can give the various input data and immediately see the result without manual calculation.

Effective presentation

Details of the Implementation:



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

The Shannon fano coding problem question was asked in Internal Assessment I (IAT I). Almost 80% of the students answered well in the examination.

Reflective Critique:

- **Benefits:**

  Students can visualize the output by changing the various input probability.

- **Challenges:**

  I planned to conduct the simulation within 30 minutes duration. Some of the students forgot the syntax of wiring program. After explaining the syntax students write the program. So it took 50 minutes to complete the demonstration.

Changes required for future (if required):

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	2	-	-	-	-	-	-	2	-	-

CO1: The Students will be able to design an application using the various source coding techniques.

References:

- S.Haykin, "Digital Communications", John Wiley, 2005
- B.Sklar, "Digital Communication Fundamentals and Applications", 2nd Edition, Pearson Education, 2009
- https://www.mathworks.com/

RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering

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<u>Innovative Practices Description</u>

UNIT II (Waveform Coding & Representation)

Degree, Semester & Branch: V Semester B.E. ECE A
Course Code & Title: EC8501 Digital Communication
Name of the Faculty member: Mr.P.Gunasekaran
Name of the Topic: Delta Modulation
Name of the Innovative Practice: Hardware Demonstration
Date & Duration: 20.07.2019 & 30 Minutes
ICT Tool Used: Communication System Trainer kit

Description:

A demonstration is the process of teaching someone how to make or do something in a step-by-step process. The benefits of demonstration as a teaching method are a better learning experience in the classroom for students; the interest in the subject; help in developing the spirit of inquiry; students cooperating in the teaching-learning process; and the process of learning becoming permanent in the student's life.

<u>Goals (Learning Outcomes):</u>

To enable the students to learn the fundamentals of delta modulation system and interpret the modulated and demodulated waveforms.

<u>Use of appropriate methods:</u>

Justification for choosing the topic using Hardware Demonstration

Using hardware demonstration students can easily understand the concepts of delta modulation systems and their output with the various input signals. They can also understand the parameters that causes the slope overload distortion and Granular noise.

<u>Effective presentation</u>

Details of the Implementation:

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

The Delta modulation question was asked in Internal Assessment II (IAT II). Almost 85% of the students answered well in the examination.

Reflective Critique:

- **Benefits:**

  Students can visualize the output waveforms by changing the various parameters of input signal.

- **Challenges:**

  I planned to conduct the hardware demonstration within 30 minutes duration. Some of the students struggled with wiring sequence. So it took 50 minutes to complete the demonstration.

Changes required for future (if required):

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	2	1	2	-	-	-	-	-	2	2	-	-

CO2: The Students will be able to interpret the various waveform coding schemes and their representation.

References:

- S.Haykin, “Digital Communications”, John Wiley, 2005
- B.Sklar, “Digital Communication Fundamentals and Applications”, 2nd Edition, Pearson Education, 2009
- http://www.anshumantech.com/Default.aspx

RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering

Academic Year: 2019- 2020 (Odd Semester)

<u>Innovative Practices Description</u>

UNIT III (Baseband Transmission & Reception)

Degree, Semester & Branch: V Semester B.E. ECE A
Course Code & Title: EC8501 Digital Communication
Name of the Faculty member: Mr.P.Gunasekaran
Name of the Topic: Eye Pattern
Name of the Innovative Practice: Lecture Video
Date & Duration: 21.08.2019 & 50 Minutes
ICT Tool Used: Computer & Projector

Description:
A video lecture is a video which presents educational material for a topic which is to be learned. It might be a video of a teacher speaking to the camera, photographs and text about the topic or some mixture of these.

<u>Goals (Learning Outcomes):</u>

To enable the students to learn the fundamentals of eye pattern and also the visual interpretation of the data rather than just numerical listing of the data.

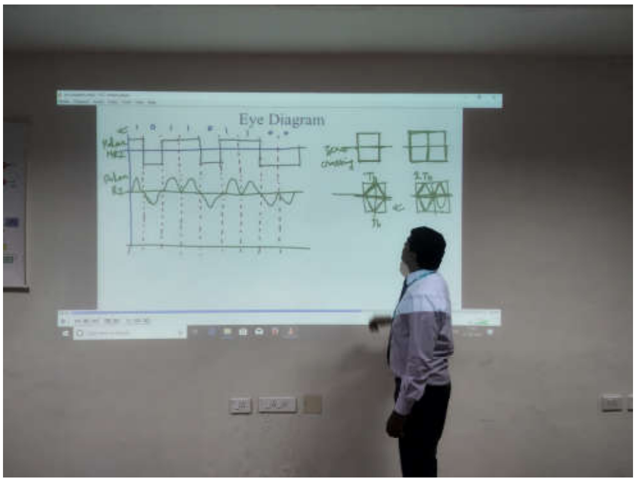
<u>Use of appropriate methods:</u>

Justification for choosing the topic using Lecture Video

The eye pattern, also referred to as the eye diagram, is produced by the synchronized superposition of successive symbol intervals of the distorted waveform appearing at the output of the receive filter prior to thresholding. From the eye pattern we can get lot of information from the pattern.

<u>Effective presentation</u>

Details of the Implementation:



Reflective Critique:

- **Benefits:**

  The students can easily gather the information concerning the degradation of quality from the eye pattern.

Changes required for future (if required):

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO3	2	2	-	-	-	-	-	-	2	1	-	1

CO3: After completion of the course, it is expected that the students will be able to analyze the various baseband transmission schemes

References:

- S.Haykin, “Digital Communications”, John Wiley, 2005
- B.Sklar, “Digital Communication Fundamentals and Applications”, 2nd Edition, Pearson Education, 2009
- http://www.digimat.in/nptel/courses/video/117104127/L52.html

RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Odd Semester)

<u>Innovative Practices Description</u>

UNIT IV (Digital Modulation Schemes)

Degree, Semester & Branch: V Semester B.E. ECE A
Course Code & Title: EC8501 Digital Communication
Name of the Faculty member: Mr.P.Gunasekaran
Name of the Topic: Quadrature Amplitude Modulation
Name of the Innovative Practice: Animation Video
Date & Duration: 18.09.2019 & 20 Minutes
ICT Tool Used: Computer & Projector

Description:

Animation is a method in which pictures are manipulated to appear as moving images. In traditional animation, images are drawn or painted by hand on transparent celluloid sheets to be photographed and exhibited on film. Today, most animations are made with computer-generated imagery (CGI). Computer animation can be very detailed 3D animation, while 2D computer animation can be used for stylistic reasons, low bandwidth or faster real-time renderings.

<u>Goals (Learning Outcomes):</u>

To enable the students to learn the fundamentals of Quadrature Amplitude Modulation system and interpret the modulated and demodulated waveforms.

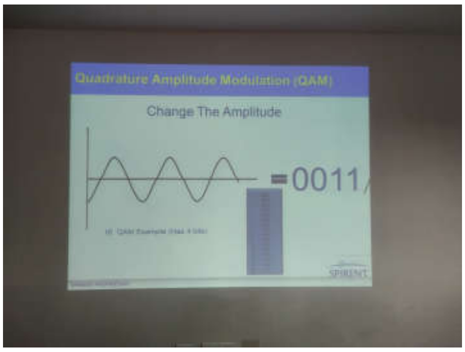
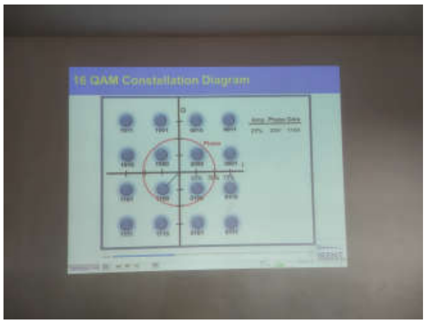
<u>Use of appropriate methods:</u>

Justification for choosing the topic using Hardware Demonstration

For the 21st generation students to compete better and to inculcate the learning process in them. Animation enables students to understand the value of reflection and critical judgment in creative work.

Effective presentation

Details of the Implementation:

Reflective Critique:

- **Benefits:**
  - Make education enjoyable
  - Inculcate critical thinking
  - analytical thinking

Changes required for future (if required):

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO4	2	2	1	-	-	-	-	1	2	2	-	1

CO4: The students will be able to develop applications using the various bandpass signalling schemes.

References:

- S.Haykin, "Digital Communications", John Wiley, 2005
- B.Sklar, "Digital Communication Fundamentals and Applications", 2nd Edition, Pearson Education, 2009
- https://www.youtube.com/watch?v=d7l5NbFfBiU

RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering

Academic Year: 2019- 2020 (Odd Semester)

<u>Innovative Practices Description</u>

UNIT V (Error Control Coding)

Degree, Semester & Branch: V Semester B.E. ECE A
Course Code & Title: EC8501 Digital Communication
Name of the Faculty member: Mr.P.Gunasekaran
Name of the Topic: Hamming Codes
Name of the Innovative Practice: Simulation Tool
Date & Duration: 26.09.2019 & 30 Minutes
ICT Tool Used: Computer and Matlab Software

Description:

MATLAB (matrix laboratory) is a multi-paradigm numerical computing environment and proprietary programming language developed by Mathworks. MATLAB allows matrix manipulations, plotting of functions and data, implementation of algorithms, creation of user interfaces, and interfacing with programs written in other languages.

<u>Goals (Learning Outcomes):</u>

To enable the students to learn the fundamentals of hamming codes and the detection and correction of errors using hamming codes.

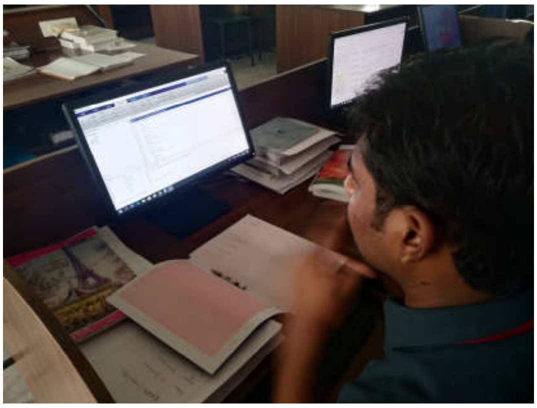
<u>Use of appropriate methods:</u>

Justification for choosing the topic using Matlab Simulation

Students manually calculate the error detection and correction capability of a coding technique for a given input binary data. The simulation can help the students to solve the problems in an effective way. They can give the various input binary data and immediately see the result without manual calculation.

<u>Effective presentation</u>

Details of the Implementation:



Reflective Critique:

- **Benefits:**
  - Implement and test your algorithms easily
  - Develop the computational codes easily
  - Debug easily
- **Challenges:**

  Some of the students forget the syntax for writing the matlab code. I helped those students and recall the syntax.

Changes required for future (if required):

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO5	3	3	2	2	-	1	-	-	-	2	-	-

CO5: The students will be able to apply the basic concepts of channel coding techniques

References:

- S.Haykin, "Digital Communications", John Wiley, 2005
- B.Sklar, "Digital Communication Fundamentals and Applications", 2nd Edition, Pearson Education, 2009
- https://www.mathworks.com/